

Lab 3: Intro to Decision Theory

Rebecca C. Steorts and Xingyu Yan

In class, you saw the resource allocation example. We will now go through how to reproduce parts of the lecture using R for all the tasks. While these will not be part of your weekly homework assignment, please do work on it on your own to make sure that you have an understanding of it.

Let's briefly recall the problem statement and set up that we saw in class.

Suppose public health officials in a small city need to decide how much resources to devote toward prevention and treatment of a certain disease, but the fraction θ of infected individuals in the city is unknown.

Suppose they allocate enough resources to accomodate a fraction c of the population. If c is too large, there will be wasted resources, while if it is too small, preventable cases may occur and some individuals may go untreated. After deliberation, they tentatively adopt the following loss function:

$$\ell(\theta, c) = \begin{cases} |\theta - c| & \text{if } c \geq \theta \\ 10|\theta - c| & \text{if } c < \theta. \end{cases}$$

By considering data from other similar cities, they determine a prior $p(\theta)$.

For simplicity, suppose

$$\theta \sim \text{Beta}(a, b)$$

(i.e., $p(\theta) = \text{Beta}(\theta|a, b)$), with $a = 0.05$ and $b = 1$.

They conduct a survey assessing the disease status of $n = 30$ individuals, $x_1, \dots, x_n$. This is modeled as

$$X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bernoulli}(\theta),$$

which is reasonable if the individuals are uniformly sampled and the population is large. Suppose all but one are disease-free, i.e., $\sum_{i=1}^n x_i = 1$.

Summary of Tasks

1. We know $p(\theta|x)$ as an updated Beta, so we can numerically compute this integral for each c . Reproduce Figure 1 from lecture, illustrating $\rho(c, x)$ for our example. Also, work through where the minimum occurs numerically ($c \approx 0.08$).
2. Now perform a sensitivity analysis for the prior assumption ($\text{Beta}(a, b)$). What do you find?
3. Consider the Bayes procedure ($c \approx 0.08$), $c = \bar{x}$, $c = 0.1$.
4. Plot the frequentist risk $R(\theta, \delta)$ as a function of θ for the three procedures in the previous task. Report your findings.
5. Based on your plot of the frequentist risk, consider the three estimators—the constant, the mean, and the Bayes estimators. By inspection of the plot alone, can you comment on admissibility versus inadmissibility of the estimators. Hint: this is a tough question, so do your best to figure this out!

Task 1

Reproduce Figure 1 from lecture, illustrating $\rho(c, x)$ for our example. Also, work through where the minimum occurs numerically ($c \approx 0.08$).

Solution:

```
# set seed
set.seed(123)

# data
sum_x = 1
n = 30
# prior parameters
a = 0.05; b = 1
# posterior parameters
an = a + sum_x
bn = b + n - sum_x
th = seq(0,1,length.out = 100)
like = dbeta(th, sum_x+1,n-sum_x+1)
prior = dbeta(th,a,b)
post = dbeta(th,sum_x+a,n-sum_x+b)
```

We now consider the loss function.

```
# compute the loss given theta and c
loss_function = function(theta, c){
  if (c < theta){
    return(10*abs(theta - c))
  } else{
    return(1 = abs(theta - c))
  }
}
```

We now write a function **posterior_risk** which is a function of c , parameters a_prior and b_prior for the prior distribution of θ , the summation of x_i sum_x , the number of observations n , and also the number of random draws s .

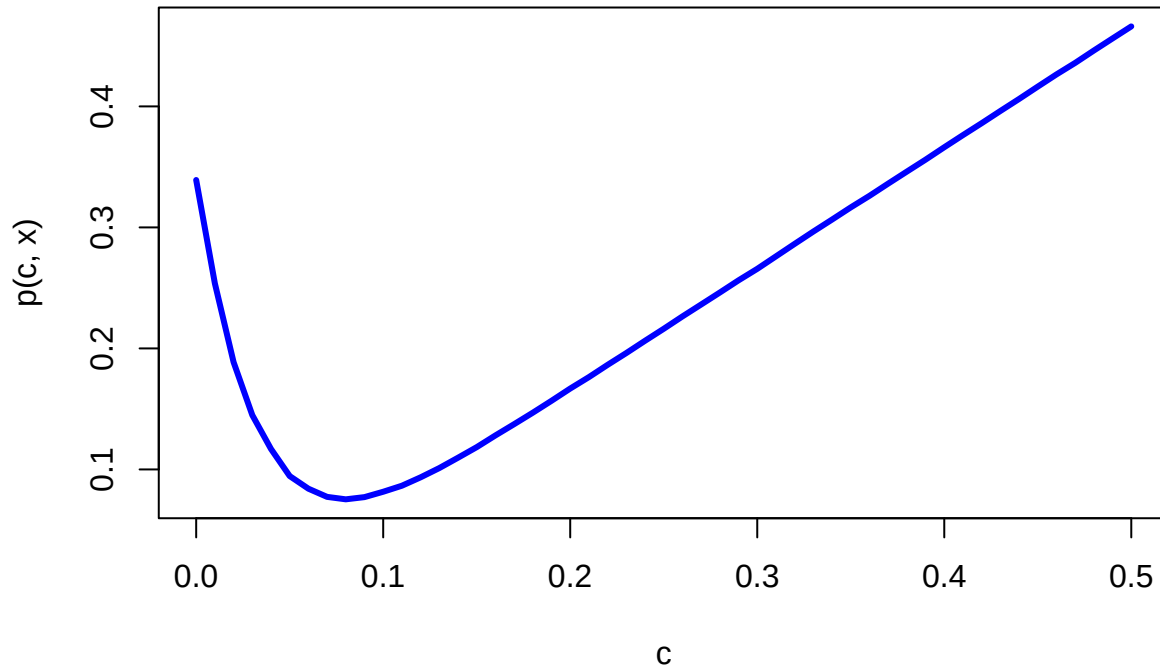
```
# compute the posterior risk given c
# s is the number of random draws
posterior_risk = function(c, a_prior, b_prior, sum_x, n, s = 30000){
  # random draws from beta distribution
  a_post = a_prior + sum_x
  b_post = b_prior + n - sum_x
  theta = rbeta(s, a_post, b_post)
  loss <- apply(as.matrix(theta),1,loss_function,c)
  # average values from the loss function
  risk = mean(loss)
}

# a sequence of c in [0, 0.5]
c = seq(0, 0.5, by = 0.01)
post_risk <- apply(as.matrix(c),1,posterior_risk, a, b, sum_x, n)
head(post_risk)
```

```
## [1] 0.33917940 0.25367603 0.18868962 0.14489894 0.11693106 0.09453471
```

We then look at the Posterior expected loss (posterior risk) for disease prevalence versus c .

```
# plot posterior risk against c
plot(c, post_risk, type = 'l', col='blue',
     lwd = 3, ylab = 'p(c, x)' )
```



```
# minimum of posterior risk occurs at c = 0.08
(c[which.min(post_risk)])
```

```
## [1] 0.08
```

We have reproduced Figure 1, and shown that the minimum of posterior risk occurs at $c = 0.08$.

Task 2

Now we perform a sensitivity analysis for the prior assumption on θ .

We set $a = 0.05, 1, 0.05$ and $b = 1, 1, 10$.

If we have different values of a, b , the posterior risk is minimized at different values of c . The optimal c depends on not only the data, but also the prior setting.

```
# set prior
as = c(0.05, 1, 0.05); bs = c(1, 1, 10)
# initialize posterior risk
post_risk = matrix(NA, 3, length(c))

# for each pair of a and b, compute the posterior risks
for (i in 1:3){
  a_prior = as[i]
  b_prior = bs[i]

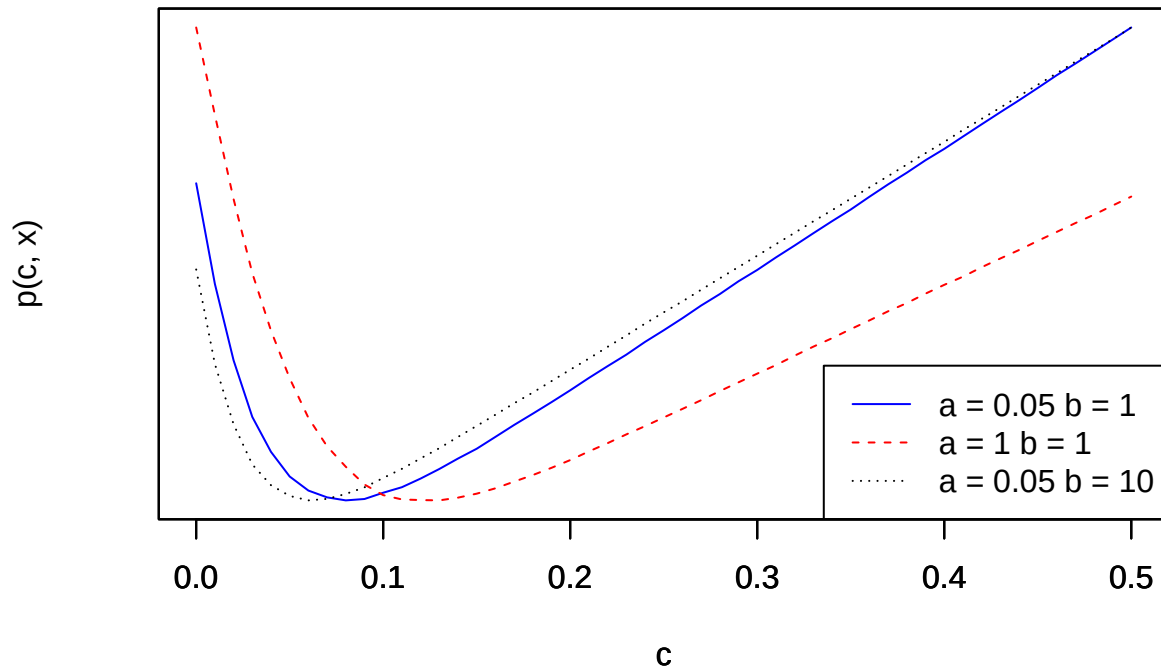
  post_risk[i,] = apply(as.matrix(c), 1, posterior_risk, a_prior, b_prior, sum_x, n)
}

# plot the posterior risk (for each prior setting)
```

```

plot(c, post_risk[1,], type = 'l', col='blue', lty = 1, yaxt = "n", ylab = "p(c, x)")
par(new = T)
plot(c, post_risk[2,], type = 'l', col='red', lty = 2, yaxt = "n", ylab = "")
par(new = T)
plot(c, post_risk[3,], type = 'l', lty = 3, yaxt = "n", ylab = "")
legend("bottomright", lty = c(1,2,3), col = c("blue", "red", "black"),
      legend = c("a = 0.05 b = 1", "a = 1 b = 1", "a = 0.05 b = 10"))

```



Remark: There is a more automated solution but this is one of most simple solutions and is completely correct.

Task 3

Consider the Bayes procedure ($c \approx 0.08$), $c = \bar{x}$, $c = 0.1$. Reproduce Figure 2. Explain your findings.

The Bayes procedure always picks c to be a little bigger than $\bar{x}$.

```

# find the sum of the x's
sum_xs = seq(0, 30)
# initialize the minimum c
min_c = matrix(NA, 3, length(sum_xs))

# find_optimal_C finds the optimal c under the Bayes procedure
# function of sum of x, parameters for prior, number of observations, and number of random draws
find_optimal_C <- function(sum_x, a_prior, b_prior, n, s = 500){
  c = seq(0, 1, by = 0.01)
  post_risk = apply(as.matrix(c), 1, posterior_risk, a_prior, b_prior, sum_x, n, s)
  c[which.min(post_risk)]
}

min_c[1,] = apply(as.matrix(sum_xs), 1, find_optimal_C, a, b, n)
# find optimal c under sample mean
min_c[2,] = (sum_xs)/n

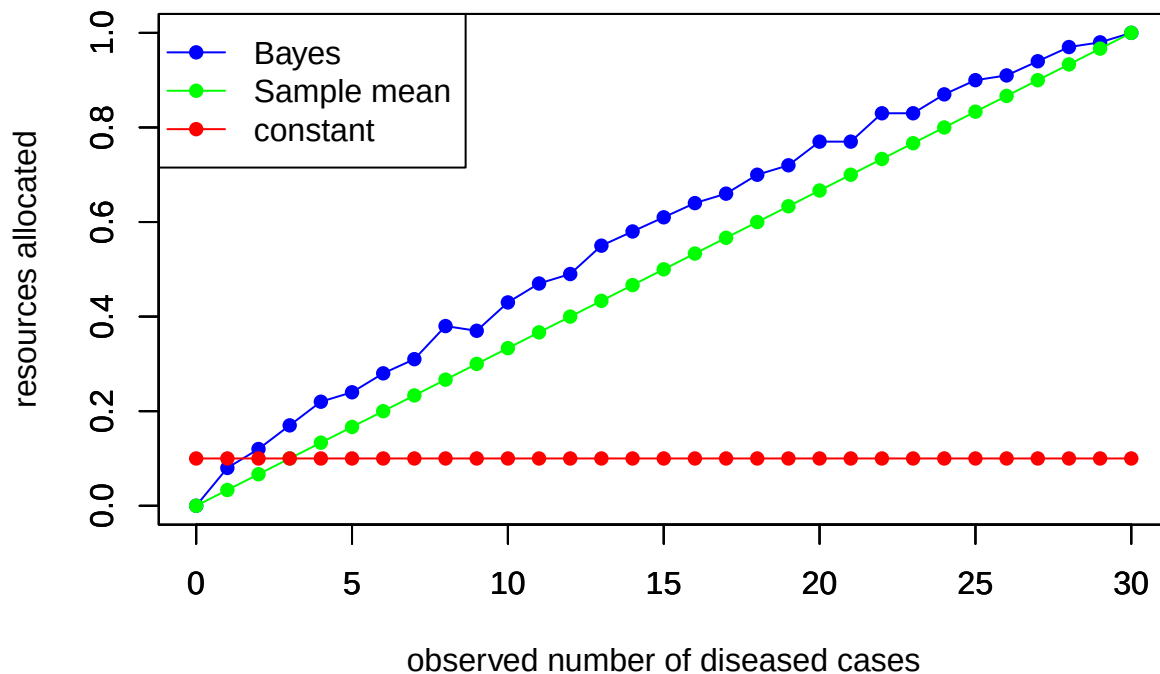
```

```

# constant c
min_c[3,] = 0.1

# plot Bayes procedure, sample mean, and constant
plot(sum_xs, min_c[1,], col='blue', type = 'o', pch = 16,
     ylab = "resources allocated", xlab = 'observed number of diseased cases',
     ylim = c(0,1))
par(new = T)
plot(sum_xs, min_c[2,], type = 'o', col='green',
     pch = 16, ylab = "", xlab = '', ylim = c(0,1))
par(new = T)
plot(sum_xs, min_c[3,], type = 'o', col = 'red',
     pch = 16, ylab = "", xlab = '', ylim = c(0,1))
legend("topleft", lty = c(1,1,1), pch = c(16,16,16),
     col = c("blue", "green", "red"),
     legend = c("Bayes", "Sample mean", "constant"))

```



Task 4

For all θ , the Bayes procedure has the lower frequentist risk than the sample mean.

```

thetas = seq(0, 1, by=0.1)

# frequentist risk for the 3 estimators given a theta
frequentist_risk = function(theta){
  sum_xs = rbinom(100, 30, theta)
  Bayes_optimal = apply(as.matrix(sum_xs), 1, find_optimal_C, a, b, n, s = 100)
  mean_c = sum_xs / 30

  loss1 = apply(as.matrix(Bayes_optimal), 1, loss_function, theta = theta)
  loss2 = apply(as.matrix(mean_c), 1, loss_function, theta = theta )
  risk1 = mean(loss1)
}

```

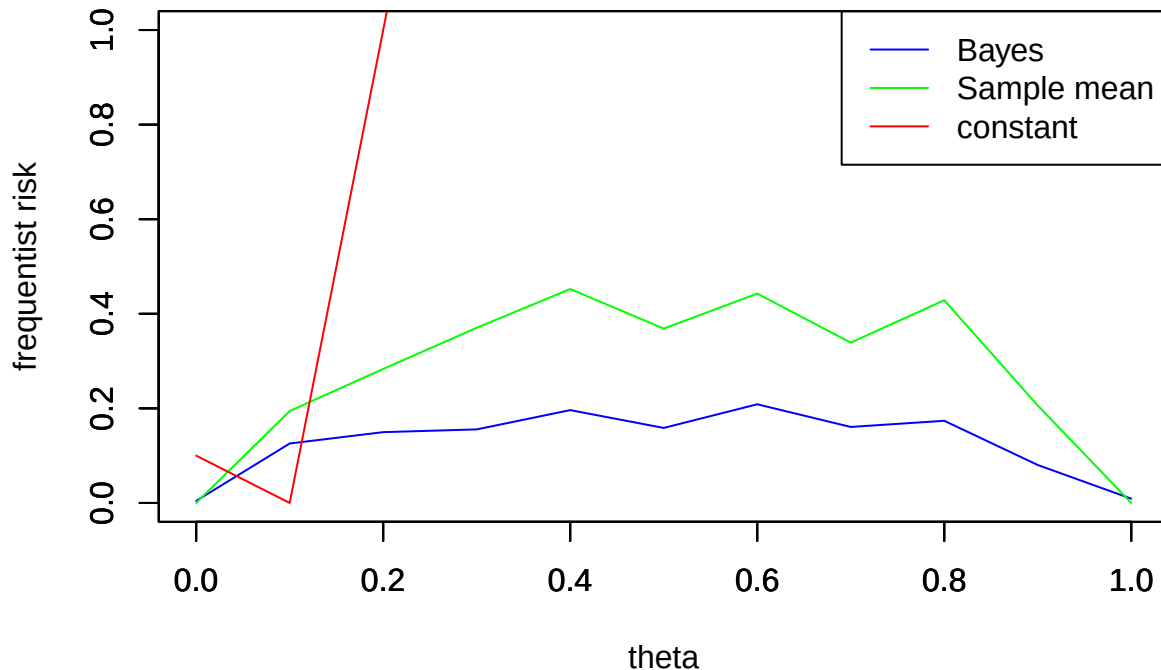
```

risk2 = mean(loss2)
risk3 = loss_function(theta, 0.1)
return(c(risk1, risk2, risk3))
}

# given a sequence a theta, compute frequentist risk for each theta
R = apply(as.matrix(thetas), 1, frequentist_risk)

# plot
plot(thetas, R[1,], col='blue', type = 'l',
     ylab = "frequentist risk", xlab = 'theta', ylim = c(0,1))
par(new = T)
plot(thetas, R[2,], type = 'l', col='green',
     ylab = "", xlab = '', ylim = c(0,1))
par(new = T)
plot(thetas, R[3,], type = 'l', col = 'red',
     ylab = "", xlab = '', ylim = c(0,1))
legend("topright", lty = c(1,1,1), col = c("blue", "green", "red"),
     legend = c("Bayes", "Sample mean", "constant"))

```



Please see a few remarks about Task 4 that will help you with interpreting the plot.

1. If you zoom into see the plot for Task 4, you will notice that the Bayes risk is not always smaller than the sample mean. Specifically, the issue is occurring around $\theta = 0$ and $\theta = 1$.
2. One observation that we can make is that when x is very small (say 0), Bayes estimator tends to overestimate θ and hence sample mean has lower risk. What other observations can you make?

I am including some code that Xu Chen has written that is much faster, where the resulting plot is more clear.

```

# code by Xu Chen
loss <- function(theta, c){
  if (c >= theta) {

```

```

    return(c - theta)
  } else {
    return(10*(theta - c))
  }
}

delta1 <- function(x){
  return(rep(0.1,length(x)))
}

delta2 <- function(x){
  return(x/30)
}

delta3 <- function(x, a = 0.05, b = 1){
  a.post <- a + x
  b.post <- b + 30 - x
  c <- seq(0,1,0.01)
  theta <- matrix(rbeta(1e4*length(c), a.post, b.post),
                  nrow = 1e4, ncol = length(c))

  return(c[which.min(sapply(c, function(u) mean(sapply(theta[,as.integer(100*u+1)], loss, c = u))))])
}

risk <- function(theta, c){
  return(sum(dbinom(x = 0:30, size = 30, prob = theta) * sapply(c, loss, theta = theta)))
}

theta.grid <- seq(0,1,0.01)
x <- 0:30

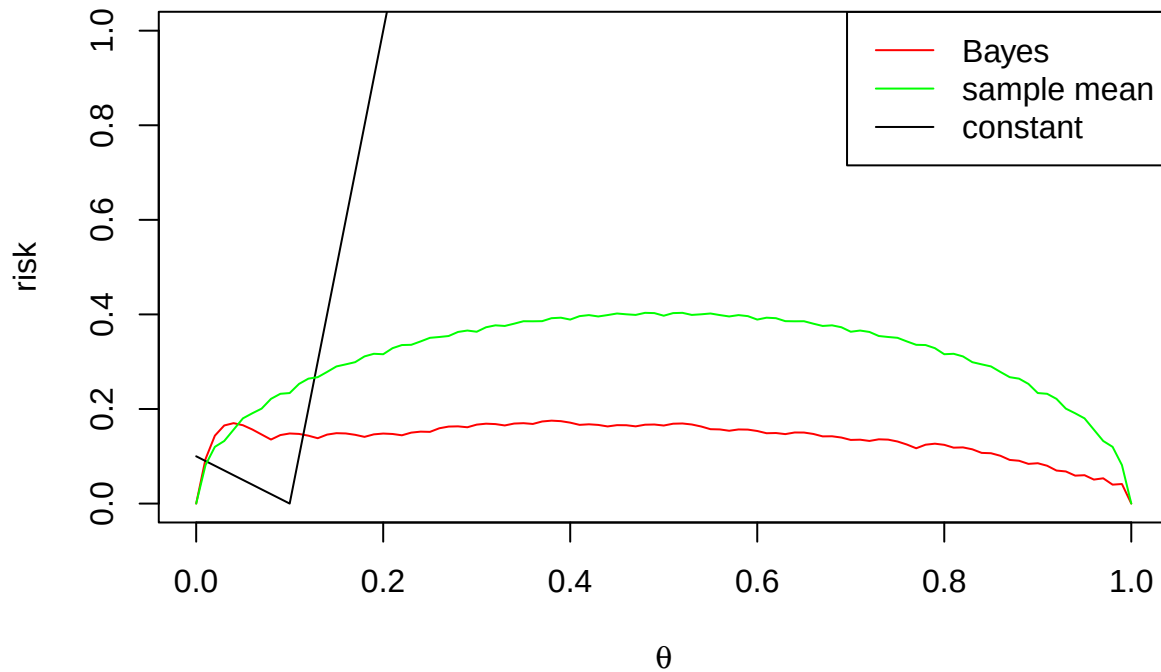
c3 <- sapply(x, delta3)
plot(theta.grid, sapply(theta.grid, risk, c3),
     ylim = c(0,1), type = 'l', col = 'red', xlab = expression(theta), ylab = 'risk')

c1 <- delta1(x)
points(theta.grid, sapply(theta.grid, risk, c1),
       ylim = c(0,1), type = 'l')

c2 <- delta2(x)
points(theta.grid, sapply(theta.grid, risk, c2),
       ylim = c(0,1), type = 'l', col='green')

legend('topright', legend = c('Bayes', 'sample mean', 'constant'),
      col = c('red', 'green', 'black'), lty = 1)

```



Task 5

Based on your plot of the frequentist risk, consider the three estimators— the constant, the mean, and the Bayes estimators. Which estimators are admissible? Be sure to explain why or why they are not admissible.

When θ is close to 0.1, the constant estimator has the smallest frequentist risk among all three estimators. However, the frequentist risk of the constant estimator increases very quickly and sharply when θ increases beyond 0.1.

The Bayes estimator has lower frequentist risk than the sample mean estimator for almost all θ values except when θ is close to 0 or 1.

When θ is close to 0 or 1, sample mean estimator has lower frequentist risk than Bayes estimator. In general, the Bayes estimator is the best estimator for most of θ values whereas the constant estimator is the best estimator for θ values close to 0.1.

Takeaway: one estimator might work better for some θ values whereas another estimator might work better for other θ values. Thus, all three estimators are admissible (no estimator is every strictly better than another estimator).